

RSA Digital Signature Scheme

Aim:

To implement Digital Signature scheme using RSA and SHA-256 in C++ for auth & non-repudiation.

Algorithm:

Key generation & Setup:

- * generate 2048-bit RSA key pair.
- * Send public key to server.

Signature Phase:

- * Hash message using SHA-256.
- * Sign digest using private key.
- * Send message & signature.

Verification phase:

- * Import public key.
- * Verify signature using EVP-DigestVerifyInit.

Output:

Client:

Generated 2048-bit RSA key pair.

Sent public key to server

Message: Authorized payment transfer request: \$5000

Generated digital signature (256 bytes)